

LLM-Powered Chatbots: A Technology Risk Assessment

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Introduction & Research Question

The main scope of this paper is to provide a comprehensive analysis and risk assessment for the implementation of Language Learning Models (LLMs), a generative AI technology that can be applied in many contexts.

By doing so, we aim to answer the following research question:

- What are the technological risks, including biases and security vulnerabilities, that are associated with the development and implementation of LLMs in multiple industries and business contexts?

LLMs can in fact be applied in many industries, and can be a digital solution at different levels, including product, system, and process levels. In order to deliver a more specific and detailed analysis, we decided to center our research on Chatbots, a specific implementation that can be differentiated from other ones (such as Predictive Tests and Virtual Assistants) in the way they are built. We further analyze how the implementation of LLM-powered Chatbots can lead to the introduction of several significant vulnerabilities, prone to being exploited by threats and ultimately leading to significant risks. After assessing this, we aim to explore what can be implemented to mitigate these risks, in order to ensure a fair and secure use of LLMs in the future.

Reference Documentation

Introduction

Our report begins with an introduction of the relevance and the contextualization of Large Language Models, which has been derived from the substantial documentation publicly available on the topic, including journalism (The Economist and others), scientific papers, and notably, a comprehensive introductory lecture on the topic made by an eminent AI scientist, Andrej Karpathy. The same lecture was also a fundamental resource in gathering information to better explain how these model work; on this topic, knowledge was integrated with the practical and insightful examples provided by Brilliant, an educational platform, and cross-checked with verifiable and trusted sources, the most relevant ones being “A Survey of Large Language Models” and “A Comprehensive Overview of Large Language Models” (2023) . Additionally, a primary resource we obtained these and other papers from is ArXiv, a Cornell-based, open-source publishing repository for scientific papers. This platform has become a hub for the latest research in the realm of LLMs and we utilized multiple documents from here throughout the report.

Business Case

For the business case, however, for Google's Bard and Salesforce's EinsteinGPT we drew information directly from their blog posts. For additional information retrieval we relied on prominent tech blog websites such as TechTarget and Zendesk. In order to draw information directly from Bank of America's implementation of their virtual assistant, we relied on PYMNTS, a prominent news outlet dedicated to research the banking industry's move towards digitalization.

Risk Assessment

Our risk assessment for Large Language Models (LLMs) primarily revolves around their cybersecurity implications. We started by consulting "A Security Risk Taxonomy for Large Language Models" from ArXiv, which details the risks associated with LLM-powered chatbots. This includes various cyberattacks and the potential for exploiting vulnerabilities. To gain deeper insight into specific types of cyberattacks, we referred to another ArXiv publication, "Multi-step Jailbreaking Attacks on ChatGPT." Additionally, we turned to mainstream media with CNBC's "ChatGPT's Jailbreak Forces it to Break its Own Rules," to understand public perception and sentiment towards the risks posed by ChatGPT. Our research also led us to an insightful paper from the National Institutes of Health (NIH), "Generative Artificial Intelligence Through ChatGPT and other Language Learning Models in Ophthalmology." Although not directly aligned with our primary focus, this paper provided an alternative perspective on the implications of LLMs in varied sectors like healthcare and education. It shed light on potential challenges, including bias, privacy concerns with confidential patient information, data handling difficulties, and interpretability issues.

Risk Mitigation

The primary resource we used for finding risk mitigation methods and best practices was OpenAI's website. As a leading research organization, OpenAI offers extensive documentation on the safe and responsible use of AI services, and publishes multiple articles on the safe use and implementation of their services. This includes strategies for reducing misinformation risks, pre-training mitigations, managing public safety risks, and best practices for the deployment of AI technologies. On the same topic, we also revisited ArXiv for additional insights, particularly the paper "Generative Language Models and Automated Influence

Operations: Emerging Threats and Potential Mitigations”, which proposes methods to counteract and adapt to the malicious use of AI and LLMs.

Additionally, we consulted an article published by Microsoft named “Planning red teaming for large language models (LLMs) and their applications”, which provided the insights necessary to implement best practices in a comprehensive way”.

What Large Language Models Are

The meteoric success achieved in 2023 by the chatbot developed by OpenAI, ChatGPT, in generating not only convincing text, but also remarkable “emergent” abilities, has brought with it concerns about the risks the implementation of Large Language Models poses, and a consequent urgency to understand how these technologies work.

Large Language Models (LLMs) are a generative AI technology that is designed to interpret human language. It is a self-improving model capable of understanding, generating, and manipulating human language via training on a large corpus of data sets, mainly containing a wide range of internet text sources. In short, LLMs acquire these abilities by using massive amounts of data to learn billions of parameters during training and consuming large computational resources in order to produce an output. (A Comprehensive Overview of Large Language Models)

These models contain two parameters that determine how the model processes and generates languages: the number of parameters in the network (n) and the amount of text that the model is trained on (D). More parameters lead to a better understanding of the language, allowing for a more advanced interpretation of different colloquialisms as well as nuances within language making them especially applicable in many different contexts and industries. (Andrej Karpathy)

Due to their vast scope of training, they are able to develop an intricate understanding of language structure, semantics, and context and can obtain more knowledge than any single human in all subjects. Additionally, LLMs are able to hear and speak, generate music, and use software infrastructure such as calculators and coding languages.

Part of the main attraction of LLMs is the models’ ability to scale quickly; the more text the models are trained on, the better the result. This drives the main motivation for the adoption of LLMs in many industries because there is a high potential for algorithmic progress at a low cost. These models are initially trained on hundreds of terabytes of internet data, then they enter the “assistant” training stage, which helps finetune the

responses of the LLM by having its training done by humans that specifically write out the optimal responses to certain questions being posed to the LLM. This allows the model to assume a type of “assistant” role when responding to queries, but also respond with large amounts of information that was obtained during the pre-training stage when the model was trained on terabytes of internet data.

Overall, the pre-training stage is about acquiring as much knowledge from training data as possible while the fine-tuning stage is about alignment - the model's ability to respond well to human interaction.

How Large Language Models work

Given a contextualization of what the technology analyzed is, it is now useful to better understand how it technically works, in order to later on evaluate what risks come with its implementation.

As a first generalization, Language Models all involve storing probabilities about which word might come next, given the preceding words. They calculate these probabilities based on sequences of words in the corpus (a collection of texts).

The simpler form of this mechanism is well represented by “Bigram Models”, algorithms that answer in an abstract way the question “given this first word, what is a likely next word?” by using only the last word provided in a prompt as a context to make probabilistic predictions. In theory, this is what is called the Markov Assumption. A more accurate predictive algorithm is the so-called “N-Gram Model”, which builds on the previously mentioned method by translating multiple words into numbers, or probabilities, but is not capable of connecting pieces of information that are separated by a lot of words. Because of this limitation, for a lot of tasks they have been replaced by more complex models such as Neural Networks, but the underlying mathematical method for counting words remains at the the core of them all.

With this in mind, one fundamental property of the more complex LLMs that can be tuned (regulated), is “Temperature”, a function that controls how random the output of a prompt is, which means its likelihood of producing novel and unexpected sequences. For example, at Temperature 0 the model eliminates all randomness from its predictions, so it chooses the most likely output every time. This type of model is called deterministic, and while on the one hand it is very reliable, on the other it does not produce the most interesting results. In practice, as a first step in creating a Large Language Model, it is necessary to turn raw text into a clean dataset in two stages: preprocessing and tokenization.

Both of these steps help standardize text in a corpus before a model is trained on it. Preprocessing involves standardizing and making decisions about how to handle variation in

language, which is necessary to make text consistent, and therefore readable for a computer. The choices made in preprocessing determine how a language model will behave later on; deciding to keep or remove capitalization and punctuation depends on the size of the corpus and the purpose of the language model.

Tokenization, instead, means dividing a corpus into smaller and more manageable units to turn text into a dataset for training large language models. This is achieved through the use of different methods, but a common drawback is that writing a program to break up words requires knowing a lot about each language's grammar. To deal with the problem of achieving similar results across different languages, there are multiple ways, but a common one is called "Byte-Pair Encoding" (BPE) and consists in building tokens from characters, which is precise enough to split the text into common tokens across languages. Once the token vocabulary has been built with BPE, it can be passed to a Neural Network for training. (Brilliant - How Large Language Models Work).

The tokens are then assigned the equivalent of definitions by embedding them into a "meaning space" where words that have similar meaning are located in nearby areas.

From here on, the LLM deploys its "attention network" to make connections between different parts of the prompt. An LLM learns these associations from scratch during its training phase - over billions of training runs, an *attention network* slowly encodes the structure of the language it sees as numbers (called "weights" within its Neural Network). In fact, LLMs have the capacity to "learn" and improve, by not only being trained on massive amounts of data, but going through the data multiple times. ("*The Generation Game*", The Economist, April 22nd 2023)

Each pass through the data is called an epoch, and during each epoch the neural network compares its prediction with the original data. There is no set number of epochs to train a model, so the ideal amount of training will differ from model to model. In doing so, the model's prediction is likely to be "off" by some amount; the difference between the predicted and actual values is called the loss, and it is an indicator to balance the model towards an optimal performance. To prevent overfitting, one effective strategy is to monitor the model's performance after each epoch once it starts performing well. This will allow you to stop training when you cease to see any improvement, or the model begins to degrade. (Brilliant - How Large Language Models Work)

Finally, once these steps are in place and the prompt has been processed, the LLM initiates a response. At this point, for each of the tokens in the model's vocabulary, the attention network has produced a probability of that token being the most appropriate one to use next in a sentence it is generating. The token with the highest probability score is not always the one chosen for the response; how the LLM makes

this choice depends on how creative the model has been told to be by its operators. (Andrej Karpahty - Intro to Large Language Models).

Business Case

LLMs are being increasingly implemented across different industries, serving various applications such as:

- Natural Language Processing and Understanding
- Chatbots and Virtual Assistants
- Content creation
- Coding assistance
- Healthcare
- Education
- Finance
- HR
- Security
- Accessibility

These applications showcase LLMs' capacity to comprehend and generate human-like text in different subjects. Focusing more on Chatbots and Virtual Assistants, they can be used for different aspects like customer support, conversational agents or virtual assistants for information retrieval.

Many companies have started to implement the use of chatbots to expedite processes and enhance customer interactions. They prove to be a smart choice for many companies because they are usually the first support interaction that customers have with the business. There are two kinds of chatbots:

- **Rule-Based Chatbots:** These chatbots are usually easy to set up and answer questions with predetermined answers.
- **AI Chatbots:** That use machine learning to understand customer requests and improve interactions.

Chatbots provide 24/7 support by reducing waiting times and providing instant assistance. They can quickly respond with information or FAQ-based responses, collect extensive customer data, and ensure a more customized experience. These include customized product recommendations, order status updates, multilingual support, easing communication barriers and ensuring a good global customer experience.

But Chatbots aren't only meant to ensure a good customer experience, but also to make the company's business smoother. AI based chatbots can learn and improve from each interaction, allowing them to track, collect, learn, and refine all the acquired knowledge to better understand the customer needs and understand whether to pass the customer to an agent or not. Through this selective mechanism, agents are able to focus their expertise on more complex issues that may not be solvable with an AI based technology. LLM based chatbots have also started to better perceive customer intent based on their language, providing additional

necessary information for an agent that can be used to match the customers with the most capable agent based on their needs.

Chatbots can also be used to do manual and time-consuming sales tasks, collecting information, qualifying leads, scheduling product demos and engaging website visitors. Focusing more on the marketing aspect, chatbots can collect consumer data while interacting with customers to understand trends and behaviors for marketing campaigns.

Chatbots serve as an effective tool for onboarding by facilitating the training process, suggesting the best possible course of action and providing easy access to information about organizational policies and frequently asked questions. All these factors contribute to streamlining the employee experience by simplifying workflows, automating, and relocating certain tasks. Additionally, chatbots offer feedback to employees and gather their insights, enabling businesses to continuously improve the employee experience. Overall chatbots will allow companies to achieve significant cost savings, increase productivity and efficiency, and handle an increasing number of requests without expanding their workforce.

There have been a few note-worthy applications of the adoption of LLM-based chatbots within different industries.

Erica: Bank of America's Virtual Assistant

Within the banking sector, Bank of America created its virtual assistant “Erica” with the intention of improving their focus towards digitization for customer growth. It is able to assist customers with cash flow, balances, transaction history, and bills. The banking sector overall prioritizes 24/7 customer support, and virtual assistants have historically been used in this sector. The difference presented through an LLM-based virtual assistant is that it provides a significant improvement in personalized experiences for every customer. In fact, after the implementation of “Erica”, Bank of America reported a significant improvement in every business segment, stating that their “consumer banking segment grew (its) sales by 4% year over year and digital sales represented 51% of total sales” (PYMN). It is clear that “Erica” has allowed Bank of America to experience significant digital growth through its improved client relationships.

Gemini: Google's Model Behind their Chatbot Bard

In the tech industry, Google developed “Gemini”, an open source AI model that powers Bard, Google's chatbot. Gemini has shown very high results, notably outperforming human experts on Massive Multitasking

Language Understanding (MMLU). This capability allows it to skillfully handle a large range of different subjects, improving its problem-solving and knowledge-application abilities. Since it uses a different benchmark approach compared to MMLU, Gemini can think more carefully in front of difficult questions. It can read, filter and understand information in many fields, from science to finance. Gemini is capable to adapt from its interactions and experiences, making it incredibly versatile and suitable for different applications, such as its integration in the Google Ads platform, providing new ways for advertisers to connect and engage with users. Its ability to evolve over time and scale ensures that it remains relevant and effective in a rapidly changing technological landscape.

Gemini's applications span multiple sectors:

- **Customer care:** Rapid contextualization of customer needs.
- **Cyber Security:** Efficient contextualization of diverse data by finding correlations between different information (IP addresses, users, devices and events) to triage security alerts for incident responses.
- **Insurance fraud detection:** Prompt review and validation of insurance claims, sales, and involved parties.
- **Audit & Compliance:** Quick resolution of authentication and access issues by answering auditor's questions with relevant user data.
- **Investment Prospecting:** Aids with the discovery of innovative companies, investors, and market competition.
- **Clinical trials research:** Contextualization of clinical trial research topics to review all relevant factors for decision making
- **Contact tracing:** Contextualization of tabular contact tracing data to identify individuals that are infected and those that are potentially at risk.

Integrating Gemini with Bard has significantly enhanced Google's chatbot-based interaction with users. Users can now experience a human-like interaction, as Bard can now respond with natural language.

Salesforce- Einstein GPT

Salesforce's Einstein GPT is an emerging AI CRM technology, by combining public and private AI models with CRM data, users can interact using natural-language prompts.

Building on this, Salesforce has launched Einstein Copilot, an AI-based chatbot designed to enhance employee efficiency in task execution. It is specifically made to assist with different business processes, offering step-by step guidance to sales associates, including strategies for closing deals. Remaining in the sales context, Einstein Copilot can be used to search specific details in customers conversations, extract next steps and action items from video calls, help employees prepare for a meeting, and automatically generate

sales emails that fit the tone and the style of the business. It can also be used to improve the customer experience, automatically answering to customers queries with personalized answers but can also assess a step-by-step guidance from the virtual assistant, ensuring a smoother workflow. For marketing initiatives, Einstein copilot is capable of generating emails, blog posts, and upgrading campaign segmentation efforts. Furthermore, Einstein Copilot extends its utility to developers by enabling them to convert natural language prompts into Apex code and identify potential vulnerabilities. This showcases its versatility in various applications. With this copilot, companies can create powerful assistants for every employee's flow of work, being able to streamline manual work, improve sales processes and revenues, and deliver superior customer experiences.

Risk Assessment

Introduction to the Risk Assessment

In the context of our topic of choice, we decided that a qualitative risk assessment would best reflect the potential risks and their qualitative potential impact and likelihood of these risks. This is because when it comes to assessing the impact of AI-based models, determining risks based on their characteristics and potential impacts can be more descriptive and subjective compared to a quantitative approach, which on the other hand may be more informative in the business context for decision making. In our case, since this is a new and emerging technology, quantitative data is limited and therefore we will present this assessment following a more general understanding of the technology.

The use of LLMs within the chatbots that businesses use can generate very realistic responses that could lead to significant implications such as the spread of disinformation, realistic impersonations, and user-interaction based harm (Derner et. al, 2023). Furthermore, LLM powered chatbots have the ability to disclose potentially sensitive information, proving to be a serious concern for organizations when it comes to data confidentiality and for individuals when it comes to data privacy.

Undoubtedly, LLM technology touches many different fields; In order to perform a detailed quantitative analysis for our risk assessment, we decided to focus our risk assessment on cybersecurity-related risks when it comes to business-implemented LLM-based chatbots, which include risks associated with data access, data

privacy, misinformation and bias, and manipulation. Furthermore, we focus on the threats that stem from the use of LLM-powered chatbots and their risks. Within these cybersecurity implications, we then classified the threats according to levels of risk on a low, medium, to high scale that considers the likelihood and impact for each threat.

Risk Determination

By just focusing on the risks within the cybersecurity context when it comes to business-implemented chatbots, possible risks can be classified within five different areas: Data Access and Privacy, Misinformation and Bias, Compliance, and most importantly the risk posed by different cyber attacks. Similarly, the risks posed by LLMs can be classified into risks related to data access and privacy, misinformation and bias, and manipulation. Many of these risks stem from adversarial attacks: crafted prompts and role-playing scenarios posed onto the chatbot.

Data Access and Privacy

Since LLMs are not only trained on data from the internet, but also are trained on a repurposed version of a user's chat history, data privacy concerns naturally stem when the data collected involves potentially confidential information. Individuals can inadvertently input sensitive information when using chatbots such as ChatGPT to assist them with work or personal matters. Individuals aren't the only culprits, enterprises that tend to outsource LLM providers to power their chatbots have had a significant shift in focus towards valuing data privacy when outsourcing sensitive data with them (Forbes). This is due to the high risk of exposure of the data in the chance that the external providers, or their personal chatbot if not outsourcing, are hacked. Additionally, organizations are predisposed to employee user error, who can insert confidential information in open-source chatbots that may contain vulnerabilities.

Jailbreaking Attacks on ChatGPT

Following ChatGPT's update to GPT 3.5 and 4, direct prompts that directly ask for sensitive data are not effective in enabling an attacker (Li, 2022). However, the model isn't foolproof when it comes to Jailbreaking prompts, which are specific prompts that side-step programming restrictions to generate anything freely. When executing this type of prompt, the model is "tricked" into responding to something that it originally should not have been able to respond to due to ethical restrictions. For example, one of the

most recent jailbreak attacks is the Do Anything Now (DAN) prompt attack. Within this prompt, the model is informed that it must act as if it is not required to abide by any rules, and to respond to any type of query bypassing these rules. This method is highly used by an attacker that aims to extract information or data that they should not have access to, violating data confidentiality.

Overall, unauthorized data access and data privacy breaches -the risks coming from a jailbreak attack- through the use of LLMs within chatbots occurs with a medium-level probability, as specific prompts and methods must be used in order to possibly reach any amount of unauthorized data. On the other hand, the impact is considered high on the occasion that it does happen because data is an extremely valuable asset and should not be compromised. Following our estimations, unauthorized data access can be considered a high level of risk, considering the importance of privacy.

Cyber Attacks - Phishing, Prompt Injection, Backdoor Attacks

The following attacks have proven to create implications when it comes to misinformation, manipulation, and bias.

Phishing

Phishing is a prominent and one of the most common attacks in cybersecurity, it is a fraudulent threat that tries to extract sensitive data from users through social engineering. Attackers have found that using chatbots such as ChatGPT improve phishing methods in various ways, one of them being generating variants of emails that can surpass safety filters that typically filter according to the same text sequences. These emails typically also have a better grammar structure, making them harder to detect (NCBI). Additionally, LLM-based chatbots can be prompted to generate malicious links disguised as hyperlinks containing reliable information. LLMs have an advanced ability to impersonate accurately a normal conversation and people, and can gather personal information from users that could be used for malicious purposes.

Phishing is a common attack, and therefore we would classify it as a high probability of occurrence by the ease of carrying out this attack using an LLM. The impact could range from being medium to high, depending on the criticality of information stolen, for the sake of clarity we classify the risk as medium.

Prompt Injection

Prompt injection attack - in the context of chatbots - is also a threat where an attacker manipulates the chatbot's input in order to cause non intentional behavior and responses. For example, a hacker might input a

prompt that is seemingly innocent, but is phrased in a way where the chatbot interprets it as a command to output something that it typically should not be doing. The risks associated with this could involve potential data leakage, the spread of false information, or overall system manipulation. Just as with phishing, a prompt injection attack contains high levels of risk due to its medium-high probability of occurrence as well as a high severity of impact.

Backdoor attacks

A backdoor attack on a chatbot has similar characteristics to prompt injection, except it focuses more on the training phase of the LLM by manipulating the model’s training environment, introducing “backdoors” that pose a number of risks. A successful backdoor attack could allow an attacker to maintain a consistent unauthorized control of the chatbot that can expand to other connected systems. If the chatbot is created by an organization, this could mean severe reputation damages if a backdoor attack is successful. Similar to the risks that stem from injection attacks, there could be a leakage of sensitive data processed by the chatbot and a manipulation of chatbot responses giving a high capability for an attacker to expand its potential control.

Overall, backdoor attacks can be considered a medium to low-level risk in terms of probability that they occur, because it is difficult to create a backdoor during the training phase (as training data is highly protected and difficult to access) and even more difficult to introduce a backdoor after it. In the off chance that a backdoor attack does occur, the risk can be considered high in terms of potential impact because it can have serious consequences on data privacy and reputation. Since we have to consider both the likelihood that it occurs (rather low) and the impact (rather high), we classify the level of risk as low-medium.

The table below summarizes the different risk levels we found associated with each type of attack:

	Jailbreak	Phishing	Prompt Injection	Backdoor
Likelihood:	Medium Likelihood	High Likelihood	Medium-High Likelihood	Low Likelihood
Impact:	High Impact	Low-Medium Impact	High Impact	High Impact
Risk:	High Risk	Medium Risk	High Risk	Low-Medium Risk

Risk Mitigation

The implications that occur from the use of LLMs with chatbots stem from three factors:

- (I) the *vulnerabilities* that are present within the model itself
- (II) *unauthorized data* access and security, and
- (III) the *lack of input validation*, which allows for multiple types of attacks.

These factors may influence one another and can overlap at certain times, so it is necessary to propose mitigation strategies and security measures that can be applied to all of them.

Enhancing existing encryption techniques and implementing stricter authentication and authorization controls can help in lowering the risk of having unauthorized data exposure.

At an organizational level, it could be in their interest to enforce stricter data handling policies especially when outsourcing an LLM for their chatbot. This can improve an organization's reputation as they are seen as more reliable. Additionally, regular security reviews can help an organization stay up to date with discovering potential vulnerabilities early.

Recommendations on how to mitigate the risks implicit to the commercial use of LLMs exist and are frequently discussed in the public debate, but are also *speculative* and expected to change significantly over time, because both the technology and accompanying safety considerations are not only new, and therefore not yet optimized, but also evolving very rapidly.

In fact, LLM limitations and avenues are being actively learned, and are expected to be updated in collaboration with the broader community over time.

Best Practices

This being said, there are several concepts that can be generalized to any model as *principles* for best-practices, and we try here to give a comprehensive assessment of them.

1. *Prohibit misuse*

To mitigate risks, publish usage guidelines and terms of use for LLMs that prohibit harmful activities, and specify high-risk domains and use cases. Also, in order to enforce these guidelines, build systems that include rate limits, content filtering, pre-approval for production access, and monitoring for unusual activity.

2. *Mitigate unintentional harm*

Risk mitigation in model development involves comprehensive evaluations to assess limitations, minimizing bias in training data, and using techniques like learning from human feedback (RLHF) to reduce unsafe and misleading behavior. It's crucial to document known weaknesses and vulnerabilities, including bias and the potential to produce insecure code, as *no preventative action* can entirely eliminate the chance for unintended harm.

3. *Thoughtfully collaborate with stakeholders*

It is generally advisable to build culturally and professionally heterogeneous teams, in order to obtain diverse perspectives and to minimize biases.

Rather than treating them as Intellectual Property, it is advisable to publicly share the security issues encountered by the private enterprises that own the models, in order to be able to make wiser decisions overall. (OpenAI)

These are to be considered as basic and generic guidelines that any Language Model is able to respect.

Regarding more specific issues, like the risk of harmful *influence operations* and/or *disinformation*, a comprehensive pathway is still applicable to the technology's risks at large.

A necessary premise here is that for malicious actors looking to spread propaganda—information designed to shape perceptions to further an actor's interest—these language models bring the promise of automating

the creation of convincing and misleading text for use in influence operations, rather than having to rely on human labor. For society, these developments bring a new set of concerns: the prospect of highly *scalable*—and perhaps even highly *persuasive*—campaigns by those seeking to covertly influence public opinion.

In terms of content dissemination, the framework included in a paper made by Georgetown University’s Center for Security and Emerging Technology, OpenAI and the Stanford Internet Observatory, suggests methods for *monitoring* and *limiting* the spread of AI-generated content, including collaboration with platforms to identify such content, requiring proof of personhood for posting, and adapting public input mechanisms to safeguard against misleading AI content.

It is equally important, however, to understand that mitigations that disrupt the supply of misleading information are ultimately only *partial* solutions if the demand for misleading information remains unchanged. While people rarely demand to be misinformed directly, information consumers often demand information that is cheap and useful for their goals—something influence operations can tailor to with greater freedom from the constraints of reality.

From a selfish perspective, moreover, ignorance is often *rational*: it is not possible to be informed on everything, gathering accurate information can be boring, and countering false beliefs may have social costs.

In fact, engaging in the spread of false information can be the result, for example, of the incentive to draw attention or boost someone’s status within a polarized social group. This is proven by the fact that when the personal costs of effortful analysis exceed the personal benefits, the likely outcome will be a low-quality contribution to group decision-making (e.g., sharing disinformation, free riding, groupthink, etc.). (Goldstein et al., 2023)

Red Teaming

As explained in the Risk Assessment, with LLMs both benign and adversarial usage can produce potentially harmful outputs, which can take many forms, including harmful content such as hate speech, incitement or glorification of violence, or sexual content.

In this regard, Red teaming is a fundamental best practice in the responsible development of systems and features using LLMs. However, it is not a replacement for systematic measurement and mitigation work, but

red teamers help to uncover and identify harms and, in turn, enable measurement strategies to validate the effectiveness of mitigations. (Microsoft)

It is also useful to:

- Test the LLM base model and determine whether there are gaps in the existing safety systems, given the context of your application.
- Identify and mitigate shortcomings in the existing default filters or mitigation strategies
- Provide feedback on failures in order to make improvements
- A best practice is to complete an initial round of manual red teaming before conducting systematic measurements and implementing mitigations. The goal of this type of practice is to identify harms, understand the risk surface, and develop the list of harms that can inform what needs to be measured and mitigated. (Microsoft)

Further Considerations and Suggestions

While instead focusing on residual risks, it is important to be reminded that many of the properties of large generative models are not fully understood, and further research is therefore necessary.

Certainly, there is still a lack of clarity regarding the *structure* and *impacts* of numerous influence operations, as they are often conducted in secrecy.

Understanding the *scale* of the threat posed by influence operations remains elusive as well.

Is the actual impact of such campaigns proportionate to the attention they receive in the popular imagination and press coverage? How effective are existing platform-based mitigations—such as *friction measures* designed to slow down the virality of content—at reducing the spread of misinformation?

Regarding influence operations using generative models, future studies should explore how these technologies affect various social groups differently. Important aspects to consider include the effectiveness of the models in different languages and the varying levels of media and internet literacy among the groups. AI developers and researchers should engage those who are likely to be affected to gain a deeper insight into their specific risks and requirements.

Several technical aspects remain uncertain as well. For example, the correlation between the size of the model, the length of fine-tuning, and its overall effectiveness or persuasiveness is still unclear. While, as said before, it's commonly accepted that larger models with extensive training excel in a wide range of tasks, including those related to disinformation, it's not certain whether *fine-tuning* a smaller model (making it more suitable for a specific task) can consistently bridge this performance gap.

How do these factors change between models primarily trained on large, well-represented languages like English and those with more capability to use less well-represented languages? On the mitigation side here, the feasibility of detection methods remains ambiguous. Although it seems reasonable to assume that

(1) attributing short pieces of content as AI-generated will remain impossible, and

(2) detection might become possible at much larger scales, it is hard to be more specific than this.

, some questions still remain:

What *scales* are necessary to enable detection? How much can perturbing models or training on *radioactive data* alter this necessary threshold? Furthermore, how realistic is it to train models in ways that reduce their likelihood of outputting misleading content to begin with?

Further research would also be useful to better understand, model, and clarify the decision-making of propagandists themselves. Detailed analyses of the *relative gains* that malicious actors can capture by incorporating generative models into their operations are also lacking. It is similarly unclear whether API (short for Application Programming Interface) restrictions on large language models meaningfully *discourage* operators from accessing certain services, and if they do, whether operators are able to simply gravitate toward open-source models without any loss of capability.

Finally, this is a rapidly moving field where *norms* have not yet solidified. Should AI developers release or restrict their models? Should internet researchers publish observed tactics of propagandists or keep them secret? To what extent can platforms and AI developers form meaningful partnerships that can aid in the detection and removal of inauthentic content?

Because of these considerations, it is therefore necessary to thoughtfully engage with the issues, both from people within the relevant industries and from neutral, third-party observers. (Goldstein et al., 2023)

Appendix

Derner, Erik, et. al. "A Security Risk Taxonomy for Large Language Models." *ArXiv*, 2023, arxiv.org/abs/2311.11415

Li, Haoran, et. al. "Multi-Step Jailbreaking Attacks on ChatGPT." *ArXiv*, 2023, arxiv.org/pdf/2304.05197.pdf.

Savoie, Christopher. "How to Mitigate the Enterprise Security Risks of LLMs." *Forbes*, Forbes Media LLC, 6 Nov. 2023, www.forbes.com/sites/forbestechcouncil/2023/11/06/how-to-mitigate-the-enterprise-security-risks-of-llms/.

Goswami, Rohan. "ChatGPT Jailbreak Forces It to Break Its Own Rules." *CNBC*, CNBC LLC, 6 Feb. 2023, www.cnbc.com/2023/02/06/chatgpt-jailbreak-forces-it-to-break-its-own-rules.html.

Tan, Ting et. al. "Generative Artificial Intelligence through ChatGPT and Other Large Language Models in Ophthalmology." *National Center for Biotechnology Information*, U.S. National Library of Medicine, 2023, www.ncbi.nlm.nih.gov/pmc/articles/PMC10598525/.

Naveed, Humza et.al. "A Comprehensive Overview of Large Language Models." *ArXiv*, 2023, <https://arxiv.org/abs/2307.06435>.

Karpathy, Andrej. "Intro to Large Language Models (1hr talk)." *YouTube*, uploaded by Andrej Karpathy November 11, 2023, www.youtube.com/watch?v=zjkBMFhNj_g.

The Economist, "Large, creative AI models will transform lives and labour markets." , 2023, <https://www.economist.com/interactive/science-and-technology/2023/04/22/large-creative-ai-models-will-transform-how-we-live-and-work>

Brilliant/Courses: "How LLMs Work", <https://brilliant.org/courses/how-llms-work/>

Wayne Xin Zhao et al., "A Survey of Large Language Models", *Arxiv*, 2023, <https://arxiv.org/abs/2303.18223>

Microsoft/Article, "Planning red teaming for large language models (LLMs) and their applications", 11/06/2023, <https://learn.microsoft.com/en-us/azure/ai-services/openai/concepts/red-teaming>

Josh A.Goldstein et al., "Generative Language Models and Automated Influence Operations:Emerging Threats and Potential Mitigations", *Arxiv*, 2023, <https://arxiv.org/abs/2301.04246>

OpenAI/Research, “Forecasting potential misuses of language models for disinformation campaigns and how to reduce risk”, 2023, <https://openai.com/research/forecasting-misuse>

OpenAI/Research, “Lessons learned on language model safety and misuse”, 2023, <https://openai.com/research/language-model-safety-and-misuse>

PYMNTS, “Bank of America’s Erica Use More Than Triples Since 2020”, 2023, <https://www.pymnts.com/earnings/2023/bank-of-america-says-all-growth-aided-by-ongoing-digitization/>

Google/The Keyword, “Introducing Gemini: our largest and most capable AI model”, 2023, <https://blog.google/technology/ai/google-gemini-ai/#sundar-note>

TechTarget, “Google Gemini”, 2023, <https://www.techtarget.com/whatis/definition/Google-Gemini>

Geminidata, “Use Cases”, 2023, <https://www.geminidata.com/use-cases/>

Zendesk Blog, “Top 22 benefits of chatbots for businesses and customers”, 2023, <https://www.zendesk.com/it/blog/5-benefits-using-ai-bots-customer-service/>

Salesforce Ben, “The Definitive Guide to Einstein GPT (Salesforce AI), 2023, <https://www.salesforceben.com/the-definitive-guide-to-einstein-gpt-salesforce-ai/#:~:text=Einstein%20GPT%20is%20a%20powerful,changing%20customer%20information%20and%20needs>

CX Today, “What is EinsteinCopilot? The Guide to Salesforce Copilot”, 2023, <https://www.cxtoday.com/wfo/what-is-einstein-copilot-the-guide-to-salesforce-copilot/#:~:text=What%20Is%20Einstein%20Copilot%20for%20Service%3F,answers%20grounded%20in%20company%20data>